🔑 **Answer Key — Teacher Copy · Lesson 5-8**

MSTAR PRACTICE WORKSHEET

Lesson 5-8: Determine Surface Area of Pyramids

Grade 6 Mathematics · Standard 6.GR.4

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: B

A square pyramid has a base edge of 5 in and a slant height of 7 in. What is the surface area?

A. 70 in²

B. 95 in²

C. 95 in³

D. 120 in²

Base: 5 × 5 = 25 in². Each lateral face: ½ × 5 × 7 = 17.5 in². Four faces: 4 × 17.5 = 70 in². SA = 25 + 70 = 95 in².

Question 2 SELECTED RESPONSE — CORRECT: B

How many lateral (triangular) faces does a square pyramid have?

A. 3

B. 4

C. 5

D. 6

A square pyramid has a square base and 4 triangular lateral faces — one for each side of the square.

Question 3 SELECTED RESPONSE — CORRECT: B

What is the area of one triangular face with base 8 in and slant height 6 in?

A. 14 in²

B. 24 in²

C. 24 in³

D. 48 in²

Area of a triangle = ½ × base × height = ½ × 8 × 6 = 24 in².

Question 4 SELECTED RESPONSE — CORRECT: C

A square pyramid has a base area of 49 cm² and a total lateral area of 84 cm². What is the surface area?

A. 49 cm²

B. 84 cm²

C. 133 cm²

D. 133 cm³

SA = Base Area + Lateral Area = 49 + 84 = 133 cm².

Question 5 SELECTED RESPONSE — CORRECT: C

Your first gift box is a square pyramid with a base edge of 7 in and a slant height of 9 in. How much wrapping paper (total surface area) covers it?

A. 126 in²

B. 144 in²

C. 175 in²

D. 175 in³

Base: 7 × 7 = 49 in². One triangular face: ½ × 7 × 9 = 31.5 in². Four faces: 4 × 31.5 = 126 in². SA = 49 + 126 = 175 in². Surface area uses square units (in²).

Question 6 SELECTED RESPONSE — CORRECT: B

Before adding the base, you need the area of just one triangular face of a gift box with base edge 8 in and slant height 5 in. What is it?

A 13 in²

B 20 in²

C 20 in³

D 40 in²

Area of a triangle = ½ × base × height = ½ × 8 × 5 = 20 in². You use the slant height (5) as the triangle's height, not the pyramid's vertical height.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: C

A square pyramid has a base measuring 6 m by 6 m and four triangular faces, each with a base of 6 m and a slant height of 5 m. What is its surface area?

A. 36 square meters

B. 60 square meters

C. 96 square meters

D. 156 square meters

Base area = 6 × 6 = 36. Each triangle = ½ × 6 × 5 = 15, and 4 × 15 = 60. Total = 36 + 60 = 96 square meters.

- **A:** That is the square base alone. Add the four triangular faces.
- **B:** That is the four triangles alone. Add the 6 by 6 base.
- **D:** The triangles were never halved: four times 6 times 5 is 120 rather than 60. A triangle is half of base times height.

Part B — correct answer: A

Why does a pyramid's surface area use the slant height rather than the vertical height?

A. The slant height is the height of each triangular face, measured along the face itself.

B. The vertical height is always longer than the slant height.

C. The slant height is easier to measure.

D. Pyramids do not have a vertical height.

Each lateral face is a triangle lying on the surface, and its height is the slant height. The vertical height runs through the inside of the solid and is not a face measurement.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that are true about the surface area of a pyramid:

A. Surface area = base area + the areas of all lateral faces.

B. A square pyramid has 4 triangular lateral faces.

C. Each lateral face area is ½ × base of the triangle × slant height.

D. The vertical height is used to find each triangular face area.

E. A pyramid's net shows the base and every lateral face laid flat.

A, B, C, and E are true. D is false — the slant height, not the vertical height, is the height of each triangular face.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

Two pyramids have the same base area (64 in²) but different slant heights. Pyramid A has slant height 6 in and Pyramid B has slant height 10 in. How much more surface area does Pyramid B have? Why does slant height affect surface area but NOT base area?

Model answer: With a 64 in² base the edge is 8 in, so Pyramid A has SA = 64 + 4(½ × 8 × 6) = 64 + 96 = 160 in² and Pyramid B has SA = 64 + 4(½ × 8 × 10) = 64 + 160 = 224 in², a difference of 64 in². Slant height only appears in the triangle formula, so it stretches the lateral faces while the base area stays 8 × 8 no matter how tall the pyramid gets.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.